



A Discounted Total Cost of Ownership and BWM-TOPSIS Framework for Value-Based Procurement of High-Cost Medical Devices in Saudi Public Hospitals

Research Proposal

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Executive Abstract

Saudi public hospitals risk buying high-cost medical devices that look affordable at tender award but become expensive once maintenance, consumables, software, training, downtime, spare parts, and disposal are capitalized over the device lifecycle. This study develops and tests a non-clinical decision-support framework for value-based procurement of high-cost medical devices in Saudi public hospitals under Vision 2030. The aim is to replace acquisition-price dominance with a transparent model that combines discounted Total Cost of Ownership (TCO) and hybrid multi-criteria decision-making.

The study uses a mixed-methods decision-analysis design. First, official Saudi policy documents, NUPCO/PIF public information, and peer-reviewed literature on TCO, value-based procurement, and MCDA will be analyzed to define context and criteria. Second, a structured expert survey targeting 15-20 healthcare finance, procurement, hospital administration, supply-chain, biomedical-engineering-management, and health-economics specialists will validate the criteria and generate Best-Worst Method (BWM) weights with consistency screening. Third, synthetic procurement scenarios will compare device alternatives using discounted TCO components: acquisition cost, installation, facility adaptation, training, lifecycle OPEX, downtime cost, software upgrades, disposal cost, and residual value. TOPSIS will rank alternatives by closeness to the ideal value-based option, and sensitivity analysis will test whether rankings remain stable under changes in criterion weights, OPEX assumptions, discount rates, downtime, and supplier-risk parameters.

Expected outputs include a weighted criteria hierarchy, discounted TCO tables, TOPSIS closeness scores, robustness tests, and a managerial dashboard for procurement committees. The contribution is not to estimate confidential Saudi prices or clinical outcomes; it is to produce a reproducible governance tool that helps hospital administrators compare affordability, service continuity, supplier reliability, digital readiness, and sustainability in one defensible procurement model. The study supports Vision 2030 by translating financial sustainability into an operational decision rule for high-cost hospital assets.

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1. Introduction and Strategic Rationale

1.1 Research Context

I treat high-cost medical-device procurement as a hospital-management decision, not as a routine purchasing exercise. Once a public hospital acquires advanced equipment, the financial burden does not stop at the tender award. Maintenance contracts, proprietary consumables, software upgrades, staff training, spare parts, downtime, facility adaptation, disposal costs, and residual-value uncertainty can turn an apparently affordable device into a weak lifecycle decision.

This problem is directly relevant to Saudi public hospitals. Vision 2030 and the Health Sector Transformation Program place financial sustainability, innovation, digital solutions, quality improvement, and integrated care at the center of health-sector reform (Health Sector Transformation Program, 2021; Saudi Vision 2030, 2026). NUPCO's unified procurement role also raises the stakes because centralized purchasing covers medicines, medical devices, and medical supplies for government hospitals and healthcare facilities (NUPCO, n.d.-a, n.d.-b). Centralization can improve standardization and spending efficiency, but it does not remove the need for device-level decision logic. Procurement committees still need a way to compare alternatives whose value cannot be reduced to acquisition price.

1.2 Problem Statement

The procurement problem is precise: Saudi public hospitals must choose among high-cost medical-device alternatives even when acquisition price, lifecycle cost, supplier reliability, service continuity, governance alignment, digital compatibility, and sustainability point in different directions.

The central dysfunction is acquisition-price dominance. A low-CAPEX device can become the more expensive choice once maintenance, consumables, software, downtime, weak local service availability, or poor interoperability are priced into the lifecycle. A higher-CAPEX alternative can be more defensible if it lowers recurrent OPEX, protects service continuity, strengthens supplier support, or aligns better with public procurement governance.

That tension creates a measurable decision problem: do procurement rankings change when alternatives are evaluated through discounted Total Cost of Ownership and weighted non-price

criteria rather than acquisition price alone? I operationalize that problem through five variables: procurement alternatives, discounted lifecycle-cost components, expert-derived criteria weights, non-price value criteria, and final alternative rankings.

1.3 Research Gap

The literature gives me useful tools, but not a complete model. Total Cost of Ownership explains why purchase price is incomplete, yet it does not fully evaluate supplier reliability, interoperability, service continuity, digital readiness, governance transparency, or sustainability. Value-based procurement broadens the meaning of value, but it often stays conceptual or clinically oriented. Multi-Criteria Decision Analysis offers weighting and ranking techniques, but generic MCDA becomes fragile when it is detached from lifecycle-cost modelling, consistency screening, and sensitivity analysis.

In the reviewed evidence, I found no sufficiently localized, non-clinical framework that integrates discounted Total Cost of Ownership, Best-Worst Method expert weighting, TOPSIS alternative ranking, and sensitivity analysis for high-cost medical-device procurement in Saudi public hospitals under Vision 2030. I do not claim that Saudi institutions lack internal procurement tools. I address a more defensible gap: the absence of a transparent, reproducible academic decision-support framework that can be tested without confidential procurement files.

1.4 Main Aim

To develop and test a non-clinical decision-support framework that integrates discounted Total Cost of Ownership and BWM-TOPSIS analysis for value-based procurement of high-cost medical devices in Saudi public hospitals under Vision 2030.

1.5 Specific Objectives

1. Identify and validate the financial, operational, governance, supplier-risk, service-continuity, digital-readiness, and sustainability criteria required to evaluate high-cost medical-device procurement alternatives in Saudi public hospitals.
2. Construct a discounted Total Cost of Ownership model that captures acquisition cost, installation, facility adaptation, training, maintenance, consumables, software upgrades, downtime cost, disposal cost, and residual value.

3. Estimate the relative importance of the selected criteria using expert judgment through the Best-Worst Method, including consistency screening and aggregation of valid expert responses.
4. Rank and stress-test synthetic procurement alternatives using TOPSIS and sensitivity analysis to determine how rankings change under alternative lifecycle-cost, discount-rate, criterion-weight, downtime, and supplier-risk assumptions.

1.6 Research Questions

Main Research Question

How can a discounted TCO and BWM-TOPSIS framework support value-based procurement of high-cost medical devices in Saudi public hospitals under Vision 2030?

Sub-Research Questions

1. Which financial, operational, governance, supplier-risk, service-continuity, digital-readiness, and sustainability criteria should be included in the evaluation of high-cost medical-device procurement alternatives in Saudi public hospitals?
2. How can acquisition cost and lifecycle OPEX components be structured into a discounted Total Cost of Ownership model for high-cost medical-device procurement?
3. How do relevant healthcare finance, procurement, hospital administration, supply-chain, and health-economics experts weight the selected criteria using the Best-Worst Method?
4. Which synthetic procurement alternative ranks highest under TOPSIS, and how robust is the ranking under alternative lifecycle-cost, discount-rate, criterion-weight, downtime, and supplier-risk assumptions?

1.7 Scope and Delimitations

I deliberately limit the study to high-cost medical-device procurement decision support in Saudi public hospitals. The unit of analysis is the procurement alternative, not the patient, disease pathway, hospital, supplier company, or national procurement system.

I include discounted lifecycle-cost modelling, expert-based criteria weighting, TOPSIS ranking, synthetic procurement scenarios, and sensitivity analysis. I focus on financial, operational, governance, supplier-risk, service-continuity, digital-readiness, and sustainability criteria.

I exclude patient records, clinical outcomes, clinical effectiveness, biomedical device performance, nursing protocols, pharmacy treatment decisions, laboratory diagnostics, internal hospital ledgers, confidential supplier bids, restricted NUPCO contract files, and binding procurement-policy design. I will not claim verified national savings or real Saudi procurement prices unless such data are publicly verifiable.

This boundary is not a weakness; it is the reason the project is executable. The thesis remains realistic within 12 months because I can build and test the decision-support framework with public documents, peer-reviewed literature, expert judgment, market-price proxies, synthetic scenarios, and sensitivity analysis.

This framing sets up the literature review: the next section tests whether existing work on TCO, value-based procurement, and MCDA is sufficient to solve the decision problem I have defined.

2. Literature Foundation / State of the Art

2.1 From Price-Based Purchasing to Lifecycle Procurement Value

Recent procurement literature exposes a weakness that matters for this proposal: acquisition price is a poor proxy for value. High-cost devices are long-lived hospital assets, and their financial impact extends beyond the purchase order into maintenance, proprietary consumables, software updates, staff training, spare parts, downtime, infrastructure adaptation, disposal, and residual value. Hinrichs-Krapels et al. (2022) show that hospitals use multiple decision processes and institutional arrangements when purchasing high-cost devices, yet evaluation can still remain fragmented and price-driven. The managerial problem, therefore, is not simply buying expensive equipment; it is choosing among alternatives whose lifecycle consequences are still uncertain at the point of procurement.

The limitation is clear. This literature identifies weaknesses in high-cost device purchasing, but it does not provide a localized decision architecture for Saudi public hospitals. Much of the evidence remains concentrated in high-income health systems and focuses on purchasing processes rather than on a reproducible model that connects lifecycle cost, supplier risk, service continuity, and governance constraints into one ranking procedure. It establishes the problem; it does not yet solve the Saudi public-hospital decision puzzle.

2.2 Total Cost of Ownership: Necessary but Not Sufficient

Total Cost of Ownership responds directly to the weakness of acquisition-price logic. TCO shifts the evaluation unit from the initial purchase price to the full lifecycle cost of owning, operating, maintaining, and retiring an asset. In medical-device procurement, this is crucial because a low-CAPEX alternative may become financially inferior once maintenance contracts, consumables, downtime, software, and disposal are included. TCO therefore provides the financial backbone of value-based procurement.

Yet TCO alone cannot answer the full procurement question. A purely financial lifecycle model can identify the alternative with the lowest discounted cost, but it cannot fully evaluate supplier reliability, local service availability, interoperability, governance transparency, digital readiness, warranty coverage, sustainability, or alignment with centralized procurement requirements. This

creates a methodological boundary: TCO improves cost visibility, but it remains incomplete when procurement value is multidimensional. The present study uses TCO not as a standalone method, but as the cost engine inside a broader multi-criteria decision framework.

2.3 Value-Based Procurement: Strong Philosophy, Weak Operationalization

Value-based procurement expands the procurement debate by arguing that medical-device decisions should consider value rather than price alone. Rahmani et al. (2021) identify VBP as a framework that considers cost, stakeholders, information, value analysis, and the weighing of evaluation criteria in medical-device procurement. MedTech-oriented VBP literature similarly argues that procurement should consider long-term value and total cost of care rather than only the initial cost of a device (Abbas et al., 2022). This supports the logic of the present study: procurement value must include more than purchase price.

Nevertheless, VBP literature often remains conceptually broad. It explains why value matters, but it does not always specify how a hospital procurement committee should mathematically translate value into ranked alternatives. In addition, VBP discussions frequently lean toward clinical or patient-outcome value. That orientation is legitimate for health technology assessment, but it is not sufficient for a non-clinical Health Services Management proposal focused on hospital financial sustainability and procurement governance. This study therefore narrows VBP into a managerial decision model: value is operationalized through lifecycle cost, expert-weighted criteria, alternative ranking, and sensitivity analysis.

2.4 MCDA in Healthcare: Transparency Requires Discipline

Multi-Criteria Decision Analysis provides the methodological bridge between lifecycle cost and multidimensional value. Healthcare decisions rarely involve one criterion; they require trade-offs among cost, risk, quality, equity, feasibility, and institutional priorities. Thokala et al. (2016) define MCDA as a structured approach for healthcare decision-making, while Marsh et al. (2016) emphasize implementation guidance and good-practice design. This literature is valuable because it shows that complex health decisions can be decomposed into criteria, weights, scores, and rankings.

However, MCDA is not automatically rigorous. If criteria are poorly defined, if expert judgments are inconsistent, if cost and benefit criteria are mixed without clear treatment, or if no sensitivity

analysis is conducted, MCDA can become quantified subjectivity. This critique matters for the present research. The study cannot simply “use MCDA”; it must design a defensible MCDA protocol with validated criteria, explicit cost/benefit orientation, consistency screening, and robustness testing. The methodological contribution therefore lies not in applying MCDA generically, but in designing a disciplined procurement-specific MCDA architecture.

2.5 Why BWM-TOPSIS Fits the Research Problem

Among weighting methods, the Best-Worst Method is suitable because it reduces the comparison burden placed on expert respondents while maintaining a consistency logic. Rezaei (2015) introduced BWM as a method in which decision-makers identify the most and least important criteria, then compare all criteria against these anchors. Compared with full pairwise comparison structures such as AHP, this is better aligned with a Master’s-level expert survey involving healthcare finance, procurement, administration, supply-chain, and health-economics professionals. The choice of BWM is therefore methodological, not cosmetic: it responds to the need for structured expert weighting under limited respondent time and possible judgment inconsistency.

TOPSIS is appropriate because the research needs to rank procurement alternatives according to their closeness to an ideal value-based option (Hwang & Yoon, 1981). The aim is not to map causal relationships among criteria; therefore, DEMATEL would add complexity without directly answering the ranking question. VIKOR could be used as a robustness comparator, but it is not necessary as the core method. TOPSIS is sufficient because it allows the study to combine cost-type and benefit-type criteria, apply BWM-derived weights, calculate distance from positive and negative ideal solutions, and produce interpretable closeness coefficients. This makes BWM-TOPSIS a coherent methodological pair: BWM answers “what matters most?”, while TOPSIS answers “which procurement alternative performs best under those weighted criteria?”

2.6 Saudi Public-Hospital Context: Why Localization Matters

The Saudi context intensifies the need for an integrated decision-support framework. Vision 2030 and the Health Sector Transformation Program emphasize financial sustainability, quality improvement, data enablement, innovation, and integrated healthcare (Health Sector Transformation Program, 2021; Ministry of Health, n.d.; Saudi Vision 2030, 2026). At the same

time, NUPCO plays a central role in unified procurement of medicines, medical devices, and supplies, with a stated focus on improving government spending efficiency (NUPCO, n.d.-a, n.d.-b, n.d.-c). This institutional context makes high-cost device procurement important: procurement decisions affect not only hospital budgets, but also public-sector accountability and service continuity.

However, centralization does not eliminate the need for decision support. Even under centralized procurement, hospitals and procurement stakeholders must still evaluate alternatives across lifecycle cost, supplier reliability, technical compatibility, service availability, and governance alignment. The reviewed literature does not provide a transparent, non-clinical, Saudi public-hospital framework that integrates these dimensions into a reproducible decision model. The Saudi context therefore does not merely provide background; it defines the need for a localized framework.

2.7 Synthesis and Gap Generation

The literature produces a clear sequence of partial solutions. First, high-cost device purchasing literature shows that acquisition-price logic is insufficient. Second, TCO literature provides a lifecycle-cost lens but cannot fully handle non-price dimensions. Third, VBP literature expands procurement value but often remains broad, conceptual, or clinically oriented. Fourth, MCDA literature provides a structured way to weight and rank criteria, but only if the method is implemented with consistency checks, transparent criteria, and sensitivity analysis. Fifth, Saudi health transformation and NUPCO's procurement role create a policy environment where a localized decision-support model is strategically relevant.

The gap is therefore practical, methodological, and contextual. In the reviewed evidence, no sufficiently localized framework integrates discounted Total Cost of Ownership, Best-Worst Method expert weighting, TOPSIS ranking, and sensitivity analysis for value-based procurement of high-cost medical devices in Saudi public hospitals under Vision 2030. This study addresses that gap by converting fragmented literature on lifecycle cost, value-based procurement, and MCDA into a non-clinical, reproducible model for hospital procurement decision support.

Transition to Section 3: The literature shows that TCO, value-based procurement, and MCDA each solve only part of the procurement problem. The next section converts these concepts into the study's analytical framework and decision criteria.

3. Conceptual Framework and Decision Criteria

3.1 Conceptual Logic of the Framework

I use an integrated decision-analysis framework rather than a purely descriptive theoretical model. The framework is built on one central proposition: value-based procurement of high-cost medical devices can be operationalized only when lifecycle cost and non-price value criteria are evaluated together. The study therefore combines three analytical lenses: Total Cost of Ownership, value-based procurement, and Multi-Criteria Decision Analysis.

Total Cost of Ownership provides the financial lens. It converts procurement from a purchase-price decision into a lifecycle-cost decision by incorporating CAPEX and future OPEX. Value-based procurement provides the managerial lens. It expands the definition of procurement value beyond cost to include supplier reliability, service continuity, digital readiness, governance alignment, and sustainability. Multi-Criteria Decision Analysis provides the decision-engineering lens. It converts heterogeneous financial and non-financial variables into weighted scores and ranked alternatives.

The framework is deliberately non-clinical. It does not evaluate patient outcomes, clinical effectiveness, biomedical superiority, or treatment pathways. The unit of analysis is the procurement alternative. Each alternative is evaluated through measurable cost and benefit criteria, weighted through expert judgment, ranked through TOPSIS, and stress-tested through sensitivity analysis.

3.2 Model Architecture

The framework follows a sequential logic:

1. Procurement alternatives are defined through synthetic high-cost device scenarios.
2. Decision criteria are selected from the literature, policy context, and expert validation.
3. Discounted TCO estimates the lifecycle financial burden of each alternative.
4. BWM weighting estimates the relative importance of each criterion using expert judgment.
5. TOPSIS ranking identifies the alternative closest to the ideal value-based procurement profile.
6. Sensitivity analysis tests whether the ranking remains stable when assumptions change.

The conceptual relationship is therefore:

Procurement alternative characteristics -> TCO and non-price criteria -> BWM criterion weights -> TOPSIS value score -> procurement ranking -> robustness classification.

This structure ensures that the framework does not merely describe procurement value; it converts procurement value into an auditable decision process.

3.3 Variables and Operational Definitions

Table 3.1 operationalizes the model variables before the analytical procedures are applied. This prevents abstract constructs such as value, governance, and service continuity from entering the framework without measurable definitions.

Table 3.1. Variables and Operational Definitions for the Procurement Decision Model

Variable type	Variable	Conceptual definition	Operational definition / measurement
Independent variable	Procurement alternative	A device/supplier/procurement option available for comparison	Synthetic alternatives defined by CAPEX, OPEX, supplier-risk, service-continuity, digital-readiness, and sustainability assumptions
Independent variable	Acquisition cost	Initial purchase cost of the device	Base-year CAPEX value assigned to each alternative
Independent variable	Lifecycle OPEX	Recurring cost of operating and maintaining the device	Discounted sum of maintenance, consumables, training, software upgrades, downtime, spare parts, and disposal cost
Independent variable	Supplier reliability	Expected ability of supplier to provide stable support	Expert-rated reliability score; warranty coverage; service-response proxy
Independent variable	Service continuity	Ability of the procurement option to minimize operational disruption	Downtime-risk score; local support availability; backup-service arrangement
Independent variable	Digital readiness	Ability to integrate with hospital information systems	Compatibility score with HIS/EHR, software integration requirement, upgrade burden
Independent variable	Governance alignment	Compatibility with transparent and centralized procurement logic	NUPCO-alignment proxy; documentation clarity; procurement transparency score
Independent variable	Sustainability	Environmental or strategic sustainability of the alternative	Waste/carbon proxy, lifecycle disposal burden, local-content alignment where applicable
Mediating variable	BWM criterion weights	Relative importance assigned to each decision criterion	Aggregated expert-derived BWM weights with consistency screening
Dependent variable	TOPSIS closeness coefficient	Overall value score of each alternative relative to ideal and non-ideal solutions	Weighted normalized score ranging from low to high relative closeness
Dependent variable	Procurement ranking	Final order of alternatives by value-based performance	Rank order generated from TOPSIS closeness coefficients

Variable type	Variable	Conceptual definition	Operational definition / measurement
Moderating variable	Sensitivity assumptions	Alternative assumptions that may change the ranking	Low/base/high scenarios for discount rate, OPEX, downtime, supplier risk, and criterion weights

Note. CAPEX = capital expenditure; OPEX = operating expenditure; BWM = Best-Worst Method; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution. The operational definitions identify how each construct will be measured through synthetic procurement scenarios, expert judgment, or sensitivity assumptions. Author's framework.

3.4 Decision Criteria

The criteria are selected because they convert the research problem into measurable procurement dimensions. Each criterion must be either cost-type, where lower values are preferred, or benefit-type, where higher values are preferred. Where local-content alignment is used, it will be treated as a strategic procurement proxy rather than a mandatory verified supplier attribute (Local Content and Government Procurement Authority, n.d.).

As summarized in Table 3.2, the decision framework converts abstract procurement value into measurable cost and benefit criteria that can be weighted through BWM and ranked through TOPSIS.

Table 3.2. Decision Criteria and Operationalization Matrix for the TCO-BWM-TOPSIS Procurement Framework

Decision dimension	Criterion	Criterion type	Operational measure	Analytical role in the model
Financial	Acquisition cost	Cost	Initial purchase price or synthetic CAPEX estimate	Captures visible upfront affordability
Financial	Installation and facility adaptation	Cost	Estimated setup, infrastructure, and installation cost	Identifies hidden implementation burden
Financial	Maintenance cost	Cost	Annual maintenance or service-contract assumption	Measures recurrent OPEX exposure
Financial	Consumables cost	Cost	Annual consumables estimate	Captures supplier lock-in and proprietary supply dependence
Financial	Training cost	Cost	Initial and recurrent staff-training estimate	Measures operational readiness cost
Financial	Software and upgrade cost	Cost	Lifecycle software, licensing, and update assumptions	Captures digital lifecycle burden
Financial	Disposal cost and residual value	Cost/Benefit	End-of-life disposal cost minus residual value	Measures terminal lifecycle impact
Operational	Downtime risk	Cost	Expected downtime probability or severity score	Captures service-disruption exposure

Decision dimension	Criterion	Criterion type	Operational measure	Analytical role in the model
Operational	Utilization suitability	Benefit	Expert-rated fit with expected hospital use intensity	Measures asset-productivity relevance
Supplier risk	Supplier reliability	Benefit	Expert score, warranty proxy, or service-performance proxy	Captures risk of supplier underperformance
Supplier risk	Warranty coverage	Benefit	Duration and scope of warranty protection	Reduces future financial and operational uncertainty
Service continuity	Local service availability	Benefit	Availability of local support and response-time proxy	Measures capacity to restore service after failure
Digital readiness	HIS/EHR interoperability	Benefit	Expert-rated compatibility with hospital information systems	Aligns procurement with digital health transformation
Governance	Procurement transparency	Benefit	Clarity of documentation and auditability of decision trail	Supports accountable procurement justification
Governance	Centralized procurement alignment	Benefit	Compatibility with NUPCO-style procurement logic	Connects hospital-level evaluation to Saudi procurement governance
Sustainability	Environmental burden	Cost	Waste, energy, disposal, or lifecycle environmental proxy	Captures sustainability-related burden
Sustainability	Local-content alignment	Benefit	Local-content or strategic alignment proxy where applicable	Links procurement value to national strategic priorities

Note. CAPEX = capital expenditure; OPEX = operating expenditure; BWM = Best-Worst Method; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; HIS = hospital information system; EHR = electronic health record; NUPCO = National Unified Procurement Company. Cost criteria are preferred when lower; benefit criteria are preferred when higher. The operational measures are designed for synthetic procurement scenarios and expert judgment; they must not be interpreted as verified Saudi procurement prices or supplier-performance data. Author's framework.

3.5 Decision Criteria and Validation Rules

The framework will be considered internally valid only if five decision rules are met.

First, each criterion must have a clear cost-type or benefit-type orientation before TOPSIS is applied. Ambiguous criteria will be revised or removed. Second, each criterion must be measurable through public sources, expert judgment, synthetic assumptions, or sensitivity parameters. Criteria requiring confidential supplier contracts or internal hospital ledgers will not be used as mandatory inputs. Third, BWM expert responses must pass basic completeness and consistency checks before aggregation. Fourth, TOPSIS rankings must be reported with

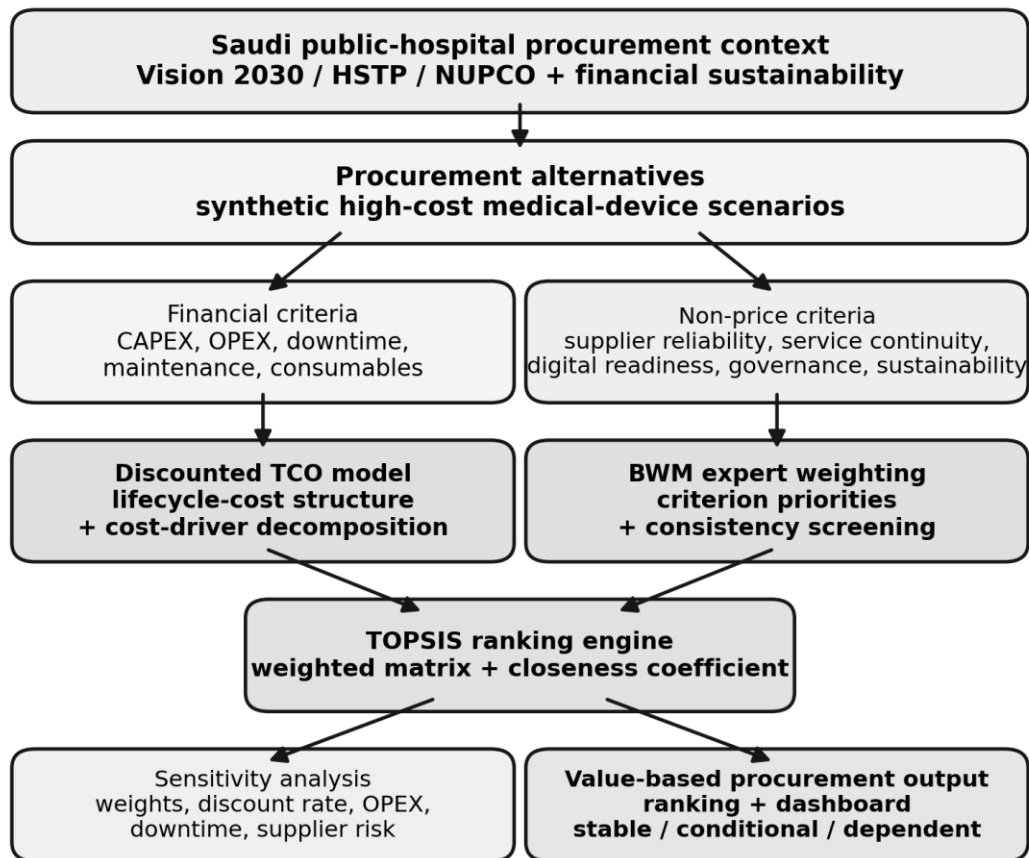
closeness coefficients, not only ordinal ranks. Fifth, sensitivity analysis must identify whether the preferred alternative is stable, conditionally stable, or assumption-dependent.

A ranking will be interpreted as stable if the same alternative remains highest-ranked across the main sensitivity scenarios. It will be interpreted as conditionally stable if the preferred alternative changes only under extreme assumptions. It will be interpreted as assumption-dependent if small changes in weights, lifecycle cost, discount rate, downtime, or supplier risk alter the preferred alternative. In the last case, the framework still provides value because it identifies which assumptions procurement committees must verify before making a decision.

3.6 Expected Visual Model

Figure 1 presents the conceptual logic of the proposed framework. It shows how Saudi public-hospital procurement context, lifecycle-cost components, expert-derived criterion weights, and non-price value criteria are converted into a transparent procurement ranking. The figure is not a decorative summary; it is the analytical map linking the research problem, the decision variables, and the final output of the study.

Figure 1. Integrated Conceptual Framework for Value-Based Procurement of High-Cost Medical Devices



Note. The framework shows the movement from procurement context to decision output. Saudi Vision 2030, the Health Sector Transformation Program (HSTP), and centralized procurement logic define the institutional context. Procurement alternatives are evaluated through financial variables, including capital expenditure (CAPEX), operating expenditure (OPEX), downtime cost, maintenance, consumables, software upgrades, disposal cost, and residual value, as well as non-price criteria such as supplier reliability, service continuity, governance alignment, digital readiness, and sustainability. Total Cost of Ownership (TCO) provides the lifecycle-cost structure. The Best-Worst Method (BWM) generates expert-derived criteria weights. TOPSIS ranks alternatives according to closeness to the ideal value-based procurement option. Sensitivity analysis tests whether the final ranking is stable, conditionally stable, or assumption-dependent.

3.7 Alignment with Research Questions and Methodology

The framework directly mirrors the research questions. The first research question identifies the criteria. The second structures CAPEX and OPEX into a discounted TCO model. The third estimates expert-derived criterion weights through BWM. The fourth ranks alternatives through TOPSIS and tests robustness through sensitivity analysis.

This alignment is essential. The conceptual framework does not introduce decorative theory; it functions as the bridge between the literature review and the methodology. It explains why the study must collect expert judgments, why lifecycle cost must be discounted, why alternatives must be ranked through TOPSIS, and why sensitivity analysis is necessary before any procurement recommendation can be considered defensible.

4. Methodology and Research Design

4.1 Methodological Positioning

I will adopt a pragmatic mixed-methods decision-analysis design. The methodology will be pragmatic because the research objective is not to test a clinical theory or estimate population-level causal effects, but to develop a usable procurement decision-support framework for high-cost medical-device purchasing under incomplete information. I will also follow a post-positivist logic because procurement value will be treated as measurable through structured criteria, explicit assumptions, expert judgment, reproducible calculations and sensitivity analysis, while recognizing that the results will remain conditional on the assumptions and expert inputs used.

I will not be designed as a clinical, biomedical, epidemiological or econometric investigation. I will not estimate patient outcomes, clinical effectiveness, treatment quality or national procurement savings. Instead, it will develop and test a non-clinical decision model that compares procurement alternatives using discounted Total Cost of Ownership, Best-Worst Method weighting, TOPSIS ranking and robustness testing.

This design is appropriate because the main research question asks how a discounted TCO and BWM-TOPSIS framework can support value-based procurement of high-cost medical devices in Saudi public hospitals. A purely qualitative design would be insufficient because the study must produce weighted criteria, lifecycle-cost calculations, rankings and sensitivity outputs. A purely quantitative econometric design would also be inappropriate because confidential supplier bids, hospital ledgers, NUPCO contracts and longitudinal device-level cost records are not publicly accessible. The proposed mixed decision-analysis design is therefore the most feasible and defensible approach for a 12-month Master's research project.

4.2 Research Design

I will conduct the research in five sequential phases.

First, a document and literature analysis will be used to define the policy context, decision criteria and methodological foundations. Official Saudi health-sector and procurement documents, peer-reviewed literature on Total Cost of Ownership, value-based procurement, MCDA, BWM and TOPSIS will be reviewed to construct the initial criteria list.

Second, the initial criteria list will be validated through expert review. Experts will assess whether the proposed criteria are relevant, clear, measurable and appropriate for high-cost medical-device procurement in Saudi public hospitals.

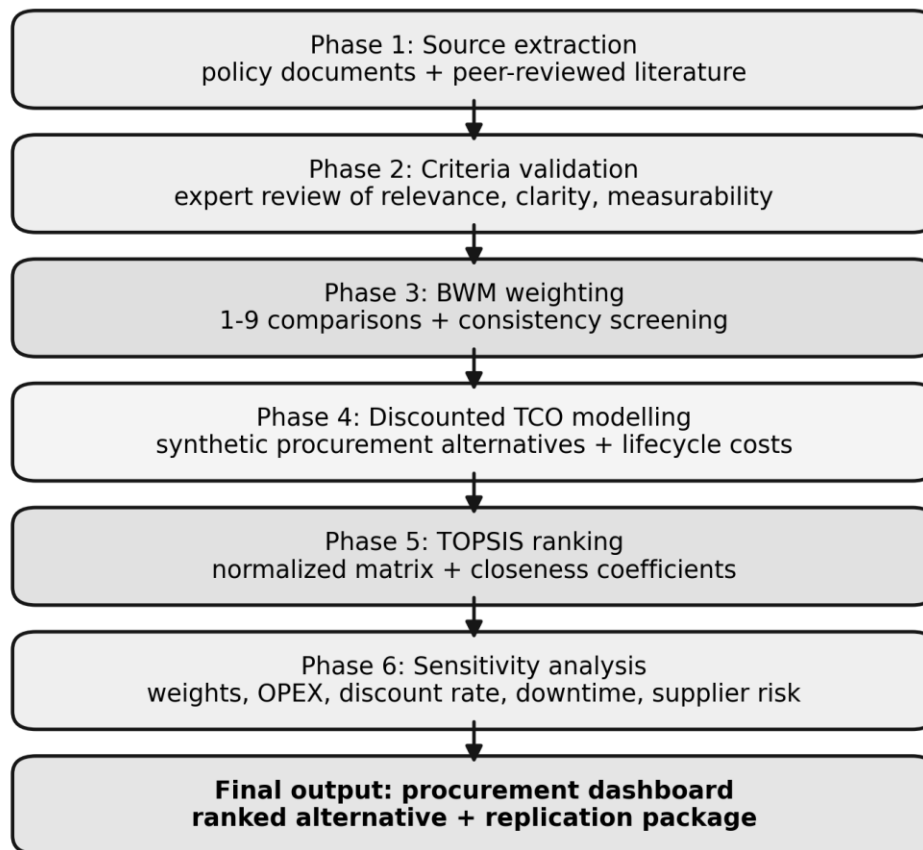
Third, the Best-Worst Method will be used to estimate criteria weights. Experts will identify the most important and least important criteria, then complete best-to-others and others-to-worst comparisons using a 1-9 preference scale. Responses will be screened for completeness and consistency before aggregation.

Fourth, a discounted Total Cost of Ownership model will be constructed for synthetic procurement alternatives. The model will include CAPEX and lifecycle OPEX components such as acquisition cost, installation, facility adaptation, staff training, maintenance, consumables, software upgrades, downtime cost, disposal cost and residual value.

Fifth, TOPSIS will be used to rank the procurement alternatives. BWM-derived weights will be applied to a normalized decision matrix containing cost-type and benefit-type criteria. Sensitivity analysis will then be conducted to determine whether the final ranking remains stable under alternative assumptions.

Figure 2 translates the research design into a reproducible methodological workflow. It clarifies how the study will move from source extraction and expert validation to BWM weighting, discounted TCO modelling, TOPSIS ranking, and sensitivity analysis. This figure allows an examiner to understand the operational sequence of the study without relying on the surrounding text.

Figure 2. Methodological Workflow from Criteria Validation to Procurement Ranking



Note. The workflow summarizes the research process. Phase 1 extracts procurement criteria from official documents and peer-reviewed literature. Phase 2 validates criteria through expert review. Phase 3 applies the Best-Worst Method (BWM) to estimate expert-derived criteria weights and screen response consistency. Phase 4 builds discounted Total Cost of Ownership (TCO) scenarios for synthetic procurement alternatives. Phase 5 applies TOPSIS to generate closeness coefficients and rank alternatives. Phase 6 conducts sensitivity analysis on criterion weights, discount rate, lifecycle OPEX, downtime cost, supplier risk, local service availability, and residual value. The workflow excludes patient records, clinical outcomes, confidential supplier bids, restricted NUPCO files, and internal hospital ledgers.

4.3 Population and Sampling Strategy

The target population will consist of experts with relevant knowledge of healthcare procurement, hospital administration, healthcare finance, biomedical equipment management, supply-chain management, health economics, public-sector governance or health technology assessment.

I will use purposive sampling because the project requires informed expert judgment, not a statistically representative survey. I will select participants for their ability to evaluate procurement criteria and make structured judgments about high-cost medical-device purchasing. This is appropriate because BWM relies on expert comparisons, not random population inference.

I will target 15-20 invitations and will require a minimum of 10 valid expert responses. The preferred valid-response range will be 12-15 experts. This range is realistic for a Master's thesis and appropriate for structured MCDA elicitation, where the goal is to obtain defensible expert weights rather than estimate population parameters.

Experts will be eligible if they meet at least one of the following criteria: professional or academic experience in healthcare procurement, hospital management, healthcare finance, medical-device purchasing, biomedical equipment management, supply-chain management, health economics, health-services management or public-sector procurement governance. A minimum of three years of relevant experience will be preferred. Respondents whose expertise is purely clinical and unrelated to procurement, finance, operations, governance or device management will be excluded.

Incomplete responses, responses with missing BWM comparisons and responses failing basic consistency or plausibility screening will not be included in the final weighting aggregation.

4.4 Data Sources

I will use six categories of data.

First, official public documents will be used to frame the Saudi policy and institutional context. These may include Vision 2030, Health Sector Transformation Program materials, NUPCO public information, Ministry of Health transformation guidance, Public Investment Fund ownership-context information where relevant, and public procurement or healthcare transformation documents (Health Sector Transformation Program, 2021; Ministry of Health, n.d.; NUPCO, n.d.-a, n.d.-b; Public Investment Fund, n.d.; Saudi Vision 2030, 2026).

Second, peer-reviewed academic literature will be used to justify the theoretical and methodological foundations of the study. Sources on TCO, value-based procurement, MCDA, BWM, TOPSIS, healthcare procurement and hospital financial sustainability will inform the criteria and analytical design.

Third, expert judgment data will be collected through a structured questionnaire. These data will support criteria validation and BWM weighting.

Fourth, synthetic procurement scenarios will be designed to test the framework without requiring confidential procurement records.

Fifth, market-price proxies and literature-based assumptions may be used to populate illustrative TCO scenarios. These proxies will not be presented as verified Saudi procurement prices.

Sixth, sensitivity assumptions will be used to test whether rankings change under alternative values for discount rate, lifecycle OPEX, downtime cost, maintenance cost, consumables cost, supplier risk and criteria weights.

I will not collect patient records, clinical outcome data, treatment data, confidential supplier bids, restricted NUPCO files, internal hospital ledgers or real contract-level procurement information.

4.5 Data Collection Procedure

Data collection will proceed in four steps.

Step 1 will consist of document and literature extraction. Relevant criteria will be extracted from official documents, academic literature and methodological sources. The extracted criteria will be grouped into financial, operational, governance, supplier-risk, service-continuity, digital-readiness and sustainability categories.

Step 2 will consist of expert questionnaire design. The questionnaire will include four parts: respondent profile, criteria validation, BWM comparisons and optional qualitative comments. The respondent-profile section will record role category, years of experience, field of expertise and exposure to procurement or hospital financial decision-making. The criteria-validation section will ask respondents to assess the relevance and clarity of each criterion. The BWM section will ask respondents to select the best and worst criteria and complete best-to-others and others-to-worst comparisons. The optional comments section will allow respondents to identify missing criteria, implementation barriers or concerns about the framework.

Step 3 will consist of pilot testing. The questionnaire will be pilot-tested with 2-3 suitable respondents to assess wording clarity, completion time, ambiguity and respondent burden. The instrument will be revised before full deployment if the pilot reveals unclear wording, overlapping criteria or excessive complexity.

Step 4 will consist of full expert survey deployment. Participants will receive an information sheet explaining the purpose of the study, expected time commitment, voluntary participation, confidentiality protections and the exclusion of confidential employer information. Responses will be collected anonymously where feasible.

4.6 Data Analysis Plan

I will analyze the data through a reproducible five-stage workflow.

4.6.1 Criteria Validation

The first stage will assess whether each proposed criterion is clear, relevant and measurable. Criteria that are conceptually overlapping, unclear or not measurable using public sources, expert judgment, synthetic assumptions or sensitivity parameters will be revised or removed. Each retained criterion will be classified as either cost-type, where lower values are preferred, or benefit-type, where higher values are preferred.

4.6.2 BWM Weighting Analysis

The second stage will estimate the relative importance of the procurement criteria using the Best-Worst Method (BWM). This method is appropriate because the study requires structured expert judgment, but it cannot impose the heavier full pairwise-comparison burden associated with larger comparison matrices. Each valid expert will identify the most important and least important criteria, then complete best-to-others and others-to-worst comparisons on a 1-9 preference scale.

To avoid treating expert preference as unexamined opinion, I will use the BWM optimization model shown in Equations (1a)-(1d). The model converts expert comparisons into normalized weights and reports the maximum inconsistency that must be minimized before the weights are used in TOPSIS.

$$\min \xi \quad (1a)$$

$$\left| \frac{w_B}{w_j} - a_{Bj} \right| \leq \xi, \quad \forall j \quad (1b)$$

$$\left| \frac{w_j}{w_W} - a_{jW} \right| \leq \xi, \quad \forall j \quad (1c)$$

$$\sum_{j=1}^n w_j = 1, \quad w_j \geq 0, \quad \forall j \quad (1d)$$

Where:

- w_j is the weight of criterion j .
- w_B is the weight of the best, or most important, criterion.
- w_W is the weight of the worst, or least important, criterion.
- a_{Bj} is the expert preference of the best criterion B over criterion j .
- a_{jW} is the expert preference of criterion j over the worst criterion W .
- n is the total number of decision criteria.
- ξ is the maximum absolute deviation minimized by the BWM model.
- j indexes the decision criteria.

Only complete and plausible BWM responses will be aggregated. The final output will include individual weights, aggregated weights, criteria rank order and consistency status. If valid

responses allow, I will also report whether criteria priorities differ across expert groups such as procurement, finance, hospital administration, supply chain and health economics.

4.6.3 Discounted TCO Modelling

The third stage will calculate the discounted Total Cost of Ownership for each synthetic procurement alternative. I will not treat the acquisition price as the full economic burden of a high-cost medical device. Instead, each alternative will be modeled as a multi-year financial commitment composed of initial CAPEX, recurrent lifecycle cost, disposal cost and residual value.

Equation (2a) defines the discounted TCO model, while Equation (2b) defines the annual lifecycle-cost component used inside the model.

$$TCO_i = CAPEX_i + \sum_{t=1}^T \frac{LC_{it}}{(1+r)^t} + \frac{DISP_i}{(1+r)^T} - \frac{RV_i}{(1+r)^T} \quad (2a)$$

$$LC_{it} = OPEX_{it} + M_{it} + C_{it} + S_{it} + D_{it} \quad (2b)$$

Where:

- TCO_i is the discounted Total Cost of Ownership of procurement alternative i .
- LC_{it} is the annual lifecycle-cost component of alternative i in year t .
- i denotes the procurement alternative being evaluated.
- t denotes the year of the lifecycle horizon.
- T is the total lifecycle horizon, in years.
- $CAPEX_i$ is the initial capital expenditure of alternative i , including acquisition, installation and facility adaptation costs.
- $OPEX_{it}$ is the annual operating expenditure of alternative i in year t .
- M_{it} is the maintenance and service-contract cost of alternative i in year t .
- C_{it} is the consumables cost of alternative i in year t .
- S_{it} is the software, licensing and upgrade cost of alternative i in year t .
- D_{it} is the downtime-related cost or operational-disruption cost of alternative i in year t .
- $DISP_i$ is the end-of-life disposal cost of alternative i .
- RV_i is the residual or resale value of alternative i at the end of the lifecycle horizon.
- r is the discount rate used as a scenario parameter.

The intuition is straightforward: a device is not cheaper because its purchase price is lower. It is cheaper only if its discounted lifecycle burden remains lower after maintenance, consumables, software, downtime, disposal and residual value are included. The TCO output will therefore report both total discounted cost and cost-driver decomposition by CAPEX, maintenance, consumables, software, downtime, disposal and residual value.

4.6.4 TOPSIS Ranking Analysis

The fourth stage will rank procurement alternatives using TOPSIS. A decision matrix will be constructed with alternatives as rows and decision criteria as columns. The matrix will include both TCO-derived financial criteria and expert-scored non-price criteria. Cost-type criteria will be oriented so that lower values are preferred, while benefit-type criteria will be oriented so that higher values are preferred.

After normalization and weighting, TOPSIS will classify each procurement alternative according to its relative closeness to the positive ideal solution and its distance from the negative ideal solution. Equation (3) gives the closeness coefficient used for the final ranking.

$$C_i^* = \frac{S_i^-}{S_i^+ + S_i^-} \quad (3)$$

Where:

- C_i^* is the TOPSIS closeness coefficient of alternative i.
- S_i^+ is the distance between alternative i and the positive ideal solution.
- S_i^- is the distance between alternative i and the negative ideal solution.
- i denotes the procurement alternative being ranked.

The coefficient C_i^* ranges from 0 to 1. A higher value means that the alternative is closer to the ideal procurement profile and farther from the worst procurement profile. The alternative with the highest C_i^* will be ranked first. The detailed normalization, weighting and distance equations are provided in Appendix VIII. The final ranking will not be interpreted as a real procurement recommendation for Saudi hospitals; it will be interpreted as a transparent test of the proposed decision-support framework under explicit synthetic assumptions.

4.6.5 Sensitivity and Robustness Analysis

The fifth stage will assess whether the rankings are robust. Sensitivity analysis will vary key assumptions one at a time and, where feasible, in combined scenarios. The tested parameters will include criteria weights, discount rate, maintenance cost, consumables cost, downtime cost, supplier reliability, local service availability and residual value.

The ranking will be classified as stable if the same alternative remains highest-ranked across the main sensitivity scenarios. It will be classified as conditionally stable if the preferred alternative changes only under extreme assumptions. It will be classified as assumption-dependent if small changes in weights or cost assumptions alter the preferred alternative. An assumption-dependent result will not invalidate the study; it will identify the variables that procurement committees must verify before making a decision.

4.7 Software and Calculation Tools

The analysis will be conducted using Excel and, where useful, Python or R. Excel will be sufficient for questionnaire coding, BWM matrices, TCO tables, TOPSIS calculations and dashboard-style outputs. Python or R may be used to automate sensitivity analysis, reproduce calculations and generate visualizations. All formulas, assumptions and calculation steps will be documented to support reproducibility.

No proprietary hospital system, confidential procurement database, clinical software or restricted institutional platform will be required.

4.8 Validity, Reliability, and Reproducibility

Construct validity will be supported by deriving criteria from literature, policy context and expert validation. Internal validity will be strengthened by documenting every analytical step from criteria selection to BWM weights, TCO calculation, TOPSIS ranking and sensitivity analysis. Reliability will be supported through a standardized questionnaire, pilot testing, clear criteria definitions, BWM consistency screening and reproducible spreadsheet or code-based calculations.

External validity will be limited because the study uses synthetic scenarios and expert judgments rather than real contract-level procurement data. This limitation is acceptable because the purpose is to develop and test a decision-support framework, not to estimate national

procurement outcomes. The framework will be designed so that authorized institutions could later replace synthetic assumptions with real internal data.

4.9 Ethical Considerations

I will involve expert judgment but will not involve patients, clinical records, medical treatment data, diagnosis, confidential supplier bids, internal hospital ledgers or restricted NUPCO contracts. Participation will be voluntary. Participants will receive information about the study purpose, expected time commitment, data use, confidentiality and right to withdraw.

No participant will be asked to disclose employer-sensitive information, non-public prices, contract terms, supplier identities or internal procurement records. Responses will be reported in aggregate form. Qualitative comments will be anonymized and stripped of identifying details. Data will be stored securely and used only for the purposes of the research.

4.10 Methodological Limitations

The first limitation is reliance on expert judgment. Expert weights may reflect professional background, organizational experience or individual bias. This will be mitigated through purposive sampling across multiple expertise areas, BWM consistency checks, aggregation of valid responses and sensitivity analysis.

The second limitation is the use of synthetic procurement scenarios. Synthetic scenarios improve feasibility and ethical safety but reduce empirical realism. This will be mitigated by making all assumptions explicit and by testing rankings under alternative cost and risk conditions.

The third limitation is the absence of confidential Saudi procurement data. I will not claim verified Saudi prices, actual supplier performance or real national savings. This limitation will be addressed by positioning the research as a framework-development and framework-testing study.

The fourth limitation is method dependence. TOPSIS rankings can be affected by normalization choices, weight structures and criteria orientation. This will be mitigated by documenting the full decision matrix, explicitly classifying cost and benefit criteria, reporting closeness coefficients and conducting sensitivity analysis.

4.11 Alignment with Research Questions

The methodology directly mirrors the research questions. Research Question 1 will be answered through literature review, document analysis and expert criteria validation. Research Question 2 will be answered through discounted TCO model construction. Research Question 3 will be answered through BWM expert weighting and consistency screening. Research Question 4 will be answered through TOPSIS ranking and sensitivity analysis.

This alignment ensures that the methodology does not generate unrelated technical outputs. Each analytical stage is designed to answer one specific research question and to contribute to the final decision-support framework.

5. Feasibility and Implementation Plan

5.1 Feasibility Logic

This project is feasible within a 12-month Master's timeline because it has been deliberately designed as a non-clinical, management-oriented, decision-analysis study. It does not require patient records, clinical outcome data, biomedical testing, laboratory access, confidential supplier bids, restricted NUPCO files, or internal hospital ledgers. The research will rely on public documents, peer-reviewed literature, expert judgment, synthetic procurement scenarios, transparent cost assumptions, and sensitivity analysis.

The operational feasibility of the study depends on four controllable conditions: timely completion of the literature and criteria framework, ethical approval for expert consultation, access to a minimum number of valid expert responses, and disciplined execution of the BWM-TCO-TOPSIS analytical workflow. I will therefore be managed as a staged research project with explicit milestones, decision gates, and contingency plans.

5.2 Project Timeline and Milestones

Table 5.1 converts the 12-month research plan into auditable phases, milestone outputs, and direct links to the study objectives.

Table 5.1. Project Timeline and Milestones

Month	Phase	Main tasks	Milestone / deliverable	Linked objective
1	Project initiation and source mapping	Finalize research scope; map Saudi policy sources, NUPCO context, TCO, VBP, MCDA, BWM, and TOPSIS literature	Approved refined outline and source map	O1
2	Literature foundation	Complete critical literature review; extract initial decision criteria	Draft Section 2 completed; preliminary criteria list created	O1
3	Framework development	Build conceptual framework; define variables; classify criteria as cost-type or benefit-type	Criteria hierarchy and conceptual model finalized	O1

Month	Phase	Main tasks	Milestone / deliverable	Linked objective
4	Ethics and instrument design	Prepare expert survey, information sheet, consent language, and ethics submission	Ethics package submitted; questionnaire draft completed	O1 / O3
5	Pilot testing	Pilot questionnaire with 2-3 respondents; revise wording, criteria definitions, and BWM instructions	Pilot completed; final survey instrument locked	O1 / O3
6	Expert recruitment	Send invitations; follow up; monitor response completeness	Minimum 10 valid responses targeted; recruitment log maintained	O3
7	BWM analysis	Clean expert data; screen completeness and consistency; calculate individual and aggregated weights	Final BWM weights and criteria ranking produced	O3
8	TCO model construction	Define synthetic procurement alternatives; build discounted TCO structure; set base-case assumptions	TCO model and cost-driver decomposition completed	O2
9	TOPSIS ranking	Build decision matrix; normalize values; apply BWM weights; calculate closeness coefficients	TOPSIS ranking matrix completed	O4
10	Sensitivity analysis	Vary discount rate, OPEX, downtime, supplier risk, residual value, and weights	Robustness classification completed	O4
11	Writing and integration	Integrate results, dashboard, contribution, limitations, and implications	Full thesis/proposal draft completed	All
12	Revision and submission	Proofread, format references, validate calculations, finalize appendices	Final submission-ready document completed	All

Note. O1-O4 refer to the four specific objectives stated in Section 1.5. The table is a planning instrument and may be adjusted if ethics approval or expert recruitment requires additional time. Author's project plan.

5.3 Gantt-Ready Work Plan

Table 5.2 provides a Gantt-ready view of the implementation plan so that the critical path can be checked visually against the methodology.

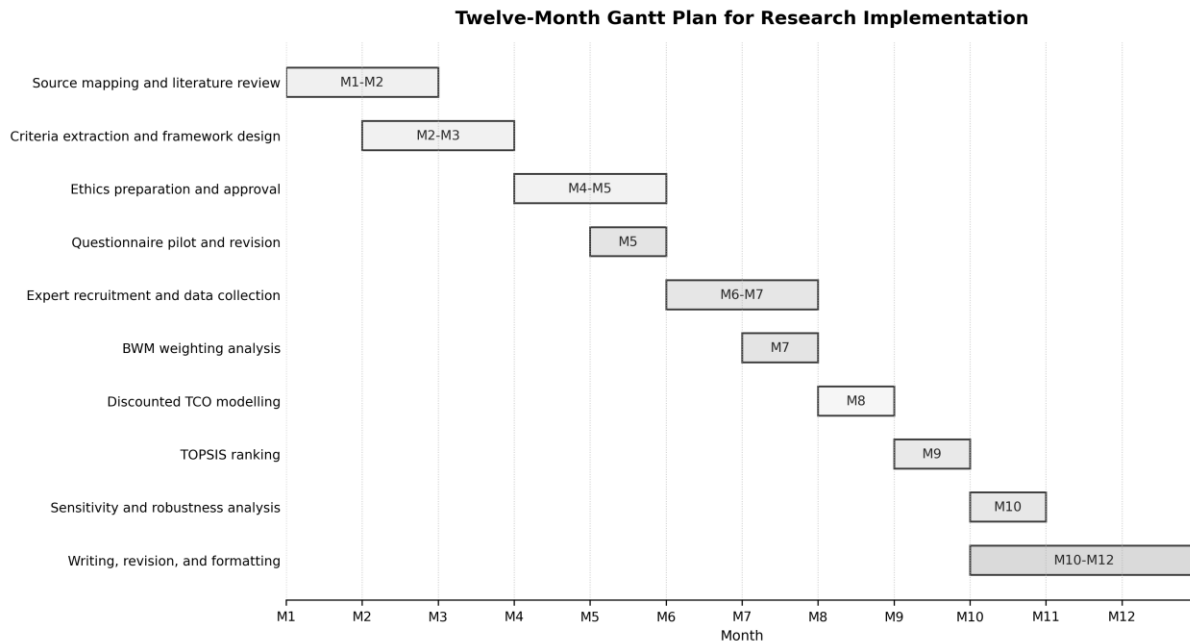
Table 5.2. Gantt-Ready Work Plan for the 12-Month Research Schedule

Work package	Active months	Milestone / deliverable	Critical-path status
Source mapping and literature review	M1-M2	Source map and literature foundation completed	Foundational
Criteria extraction and framework design	M2-M3	Criteria hierarchy and conceptual model finalized	Foundational
Ethics preparation and approval	M4-M5	Ethics package submitted or cleared; questionnaire ready	Critical path
Questionnaire pilot and revision	M5	Pilot completed; final survey instrument locked	Critical path
Expert recruitment and data collection	M6-M7	Minimum 10 valid expert responses targeted; recruitment log maintained	Critical path
BWM weighting analysis	M7	Aggregated BWM weights and consistency screening completed	Critical path
Discounted TCO modelling	M8	TCO model and cost-driver decomposition completed	Critical path
TOPSIS ranking	M9	Closeness coefficients and ranking matrix completed	Critical path
Sensitivity and robustness analysis	M10	Robustness classification completed	Critical path
Writing, revision, and formatting	M10-M12	Full draft, final revision, formatting, and submission completed	Completion

Note. M = project month. The table summarizes the same work packages shown visually in Figure 3 but adds explicit milestones and critical-path status. Author's project plan.

Figure 3 demonstrates the operational feasibility of the project by converting the methodology into a 12-month implementation plan. The figure links each work package to a concrete milestone, ensuring that the proposal is evaluated as an executable Master's research project rather than as an open-ended conceptual exercise.

Figure 3. Twelve-Month Gantt Plan for Research Implementation



Note. The Gantt plan organizes the project into sequential work packages: source mapping, literature review, criteria extraction, conceptual framework design, ethics submission, questionnaire development, pilot testing, expert recruitment, BWM weighting, discounted TCO modelling, TOPSIS ranking, sensitivity analysis, writing, revision, and final submission. The critical path is ethics approval, expert recruitment, BWM weighting, and completion of the TCO-TOPSIS analysis. The minimum viable project requires a validated criteria hierarchy, at least 10 valid expert responses, three synthetic procurement alternatives, a discounted TCO model, TOPSIS rankings, sensitivity analysis, and a managerial dashboard.

5.4 Resource and Budget Requirements

The study has been designed as a low-cost project. No laboratory, clinical equipment, proprietary hospital database, field travel to hospitals, or paid commercial procurement dataset is required. The budget below is an operational planning budget. It is not a claim of confirmed institutional funding.

Table 5.3 specifies the minimum resources required to execute the methodology without laboratory infrastructure, confidential procurement databases, or proprietary clinical systems.

Table 5.3. Resource and Budget Requirements

Budget item	Purpose	Estimated cost	Cost-control strategy
Literature access	Academic articles, reports, and policy documents	SAR 0.00-500.00	Use university library access, open-access articles, official Saudi sources, WHO/OECD/World Bank sources where available
Survey platform	Expert questionnaire distribution	SAR 0.00	Use Google Forms, Microsoft Forms, or institutional survey access
Reference management	Citation management and APA formatting	SAR 0.00	Use Zotero or equivalent free reference manager
Quantitative analysis tools	BWM, TCO, TOPSIS, sensitivity analysis	SAR 0.00	Use Excel plus R or Python where needed
Data storage	Secure storage and backup of anonymized survey files	SAR 0.00-150.00	Use institutional cloud storage or encrypted local backup
Expert recruitment	Email outreach and follow-up	SAR 0.00	Use professional/academic networks and public contact channels
Participant appreciation	Optional symbolic thank-you or certificate, if permitted by ethics rules	SAR 0.00-500.00	Avoid financial incentives unless approved and necessary
Communication	Online calls or clarification meetings	SAR 0.00-200.00	Use free video-conferencing and email
Contingency reserve	Unexpected formatting, translation, data-cleaning, or software needs	SAR 500.00	Keep reserved only for documented research needs
Estimated total		SAR 500.00-1, 850.00	Low-cost, software-light design

Note. SAR = Saudi riyal. Amounts are operational planning estimates, not confirmed funding awards. The budget assumes use of free or institutionally available tools where possible. Author's budget estimate.

The budget is proportionate to the methodology. The project requires analytical discipline rather than expensive infrastructure. R, Python, and Zotero provide no-cost or open-source support for statistical computing, reproducible calculation, and reference management. Google Forms or Microsoft Forms can be used for basic expert survey administration.

5.5 Data Access Feasibility

The project is feasible because it does not depend on restricted institutional datasets. The minimum viable dataset consists of three elements: a validated criteria list, at least 10 valid expert responses for BWM weighting, and synthetic procurement alternatives populated with transparent TCO assumptions. These inputs are sufficient to test the decision-support framework.

I will not require access to confidential supplier contracts, real NUPCO tender files, internal hospital cost ledgers, or patient-level datasets. This restriction is deliberate. It reduces ethical risk, shortens the approval pathway, and makes the project executable within a Master's timeline. If authorized institutions later provide real procurement data, the framework can be populated with real values; however, the feasibility of the thesis will not depend on such access.

5.6 Minimum Viable Thesis Design

The minimum viable version of the study will be completed if the following outputs are delivered:

Table 5.4 defines the minimum viable thesis design so that the project remains executable even if access to real procurement data is not granted.

Table 5.4. Minimum Viable Thesis Requirements

Requirement	Minimum acceptable output
Literature and policy foundation	Critical review linking TCO, VBP, MCDA, BWM, TOPSIS, HSTP, and NUPCO context
Criteria framework	Final criteria hierarchy with cost/benefit classification
Expert weighting	Minimum 10 valid expert responses and aggregated BWM weights
TCO modelling	At least three synthetic procurement alternatives with discounted lifecycle-cost assumptions
Ranking analysis	TOPSIS closeness coefficients and rank order
Robustness analysis	Sensitivity tests on weights, discount rate, OPEX, downtime, and supplier risk
Managerial output	Procurement dashboard or decision matrix interpretable by hospital administrators

Note. TCO = Total Cost of Ownership; VBP = value-based procurement; MCDA = Multi-Criteria Decision Analysis; BWM = Best-Worst Method; TOPSIS = Technique for Order Preference by Similarity to Ideal Solution; HSTP = Health Sector Transformation Program; NUPCO = National Unified Procurement Company. Author's feasibility framework.

5.7 Risk Management and Contingency Plan

Table 5.5 identifies the main execution risks and the contingency actions that will protect the study if ethics approval, expert recruitment, cost assumptions, or ranking stability become problematic.

Table 5.5. Risk Register and Contingency Plan

Risk	Probability	Impact	Early warning signal	Mitigation / Plan B
Ethics approval is delayed	Medium	High	No approval by Month 5	Prepare ethics documents in Month 4; use anonymous expert survey; avoid sensitive employer data; continue literature, TCO model, and scenario design while awaiting approval
Insufficient expert responses	Medium	High	Fewer than 10 valid responses by mid-Month 7	Expand recruitment to academics, healthcare finance professionals, procurement specialists, supply-chain experts, biomedical equipment managers, and health economists; extend data collection by 2-3 weeks; simplify questionnaire if burden is high
BWM responses are inconsistent	Medium	Medium	Multiple responses fail consistency or completeness checks	Exclude invalid responses; provide clearer BWM instructions; run a second clarification round if ethically permitted; use sensitivity analysis to test weight instability
Public cost data are weak	High	Medium	Large uncertainty around market-price proxies	Treat cost inputs as synthetic scenario assumptions; use low/base/high values; avoid claiming real Saudi prices; emphasize framework testing rather than price estimation
TOPSIS ranking changes under small assumption shifts	Medium	Medium	Preferred alternative changes during sensitivity testing	Report result as assumption-dependent; identify critical variables that procurement committees must verify before decision-making

Risk	Probability	Impact	Early warning signal	Mitigation / Plan B
Scope creep	Medium	High	Addition of clinical outcomes, real contracts, or national savings claims	Enforce exclusions: no clinical outcomes, no confidential data, no verified savings claims; keep the unit of analysis as procurement alternative
Software or technical difficulty	Low	Medium	Calculation errors or inability to automate analysis	Use Excel as baseline; use Python/R only where useful; document formulas manually; cross-check calculations on sample data
Timeline slippage	Medium	High	BWM analysis not complete by Month 8	Freeze criteria list; reduce optional subgroup analysis; prioritize core outputs: weights, TCO, TOPSIS, sensitivity

Note. Probability and impact are qualitative risk ratings. The contingency plan prioritizes preservation of the core outputs: criteria hierarchy, BWM weights, discounted TCO model, TOPSIS ranking, and sensitivity analysis. Author's risk register.

5.8 Operational Alignment with Objectives

Table 5.6 tests whether each planned task is connected to a stated objective, an implementation window, and a tangible output.

Table 5.6. Operational Alignment Between Objectives, Tasks, Timeline, and Outputs

Objective	Operational task	Timeline	Output
O1: Identify and validate criteria	Literature extraction, policy review, expert validation	M1-M5	Final criteria hierarchy
O2: Construct discounted TCO model	Define cost components, lifecycle assumptions, and discounting structure	M8	TCO model and cost decomposition
O3: Estimate criteria weights	Expert survey, BWM calculation, consistency screening	M6-M7	Aggregated BWM weights
O4: Rank and stress-test alternatives	TOPSIS matrix, closeness coefficients, sensitivity analysis	M9-M10	Ranking and robustness report

Note. O1-O4 refer to the four specific objectives in Section 1.5. The table is used to prevent orphan tasks that are not required by the research questions or methodology. Author's alignment matrix.

No planned task is orphaned. Each work package directly supports one research objective and one analytical output.

5.9 Feasibility Conclusion

This project is operationally feasible because it is low-cost, non-clinical, software-light, and independent of restricted procurement databases. Its main risk is expert access, not infrastructure or budget. That risk is controlled through purposive sampling, a minimum viable response threshold, contingency recruitment, and sensitivity analysis. The project is therefore realistic for a 12-month Master's research timeline and remains aligned with the methodology, research questions, and expected contribution.

6. Expected Outputs and Contributions

6.1 Expected Deliverables

I will produce a concrete decision-support package for value-based procurement of high-cost medical devices in Saudi public hospitals. I will not claim verified national cost savings, real Saudi procurement prices, or actual supplier performance unless such data are publicly verifiable. The contribution is deliberately precise: a reproducible analytical framework that shows how procurement alternatives can be compared when acquisition price, lifecycle cost, operational risk, governance alignment, digital readiness, and sustainability lead to competing conclusions.

The expected deliverables are:

1. **A validated procurement criteria hierarchy** covering financial, operational, governance, supplier-risk, service-continuity, digital-readiness, and sustainability dimensions.
2. **A discounted Total Cost of Ownership model** capturing acquisition cost, installation, facility adaptation, training, maintenance, consumables, software upgrades, downtime cost, disposal cost, and residual value.
3. **A BWM criteria-weighting structure** showing individual and aggregated expert weights, criteria ranking, and consistency status.
4. **A TOPSIS procurement-ranking matrix** comparing synthetic procurement alternatives through weighted cost and non-cost criteria.
5. **A sensitivity-analysis report** testing whether procurement rankings remain stable under alternative discount-rate, OPEX, downtime, supplier-risk, residual-value, and weighting assumptions.
6. **A managerial procurement dashboard** translating the model into an interpretable tool for procurement committees and hospital administrators.
7. **A replication package** containing documented assumptions, formulas, criteria definitions, and calculation workflow so that authorized institutions can later replace synthetic values with real internal data.

6.2 Theoretical Contributions

The theoretical contribution is not the invention of a new mathematical algorithm. It is the disciplined integration of fragmented literatures into a single health-services-management framework.

First, I extend Total Cost of Ownership literature by embedding lifecycle costing inside a multi-criteria procurement model. TCO explains why acquisition price is incomplete, but it does not resolve trade-offs involving supplier reliability, service continuity, governance, digital readiness, or sustainability. I position TCO as the financial engine of a broader decision framework rather than as a standalone costing exercise.

Second, I operationalize value-based procurement in a non-clinical hospital-management context. Much of the value-based procurement literature emphasizes broad value principles or clinical outcomes. I translate value into measurable administrative variables: discounted lifecycle cost, OPEX exposure, downtime risk, supplier support, governance alignment, digital compatibility, and sustainability proxies.

Third, I contribute to MCDA practice in healthcare procurement by combining BWM weighting, TOPSIS ranking, and sensitivity analysis in a transparent sequence. This matters because MCDA becomes weak when criteria are poorly defined, expert judgments are unchecked, or rankings are reported without robustness testing. The framework makes criteria selection, weighting, ranking, and stress-testing explicit.

Fourth, the study contributes to Saudi health-services-management research by localizing procurement decision support within the context of Vision 2030, the Health Sector Transformation Program, and centralized procurement governance. The gap addressed is therefore methodological and contextual: a Saudi public-hospital model that integrates discounted TCO, expert-derived weights, alternative ranking, and robustness testing for high-cost medical-device procurement. This administrative and hospital-management orientation is also consistent with the target program's focus on healthcare services and hospitals management (King Abdulaziz University, n.d.).

6.3 Practical and Societal Impact

The practical value of the study lies in improving the transparency and defensibility of procurement reasoning. Hospital administrators and procurement committees often face alternatives that cannot be judged by acquisition price alone. A lower-CAPEX device may carry higher maintenance cost, consumables dependence, software burden, downtime risk, or weak local service support. Conversely, a higher-CAPEX option may be more defensible if it produces lower lifecycle exposure and stronger service continuity.

For healthcare finance managers, the discounted TCO model will provide clearer visibility over multi-year cost exposure. It will help distinguish visible acquisition cost from hidden lifecycle liabilities.

For procurement and supply-chain professionals, the BWM-TOPSIS model will provide a structured way to justify why one alternative ranks higher than another. Each ranking will be traceable to explicit criteria, expert weights, cost assumptions, and sensitivity tests.

For public-sector decision-makers, the framework will support auditability. I will not replace procurement judgment, but it will make procurement reasoning easier to review, challenge, and improve.

For society, the value is indirect but important: stronger procurement decisions can support better stewardship of public healthcare resources. I will not claim direct patient-outcome improvement, but it will address one managerial condition that affects hospital sustainability: the ability to avoid short-term purchasing decisions that create long-term financial and operational burden.

6.4 Five-Year Value Horizon

Within five years, this type of framework could support three realistic developments.

First, hospital procurement teams could use standardized lifecycle-cost templates before awarding high-cost device contracts. Second, procurement committees could apply structured multi-criteria scoring to justify decisions where the cheapest option is not the strongest lifecycle-value option. Third, authorized institutions could populate the framework with real internal procurement, maintenance, downtime, and supplier-performance data, transforming the model from a synthetic research tool into an institutional decision-support instrument.

The expected long-term value is therefore not a universal ranking of all medical devices. It is the creation of a decision architecture that can be reused, adapted, and strengthened as better data become available.

6.5 Dissemination Plan

I will not be treated as library-only research. Its outputs will be designed for academic and practitioner dissemination.

The academic dissemination plan will include a thesis manuscript, a conference-style research presentation, and a journal article draft targeting healthcare management, health economics, procurement, or MCDA-oriented outlets. The article will focus on the integration of discounted TCO, BWM, TOPSIS, and sensitivity analysis under restricted data conditions.

The practitioner dissemination plan will include an executive brief for hospital administrators and procurement professionals, a one-page decision-framework summary, and a spreadsheet-based prototype of the procurement dashboard. Where ethically and institutionally permissible, the calculation template and anonymized synthetic example may be shared as a reproducibility appendix or open educational tool.

The policy-facing dissemination plan will include a concise policy note explaining how lifecycle costing and structured multi-criteria evaluation can strengthen procurement governance without requiring access to confidential supplier contracts or patient data.

6.6 Contribution-to-Gap Alignment

The research gap identified in this proposal is the absence, in the reviewed evidence, of a localized, non-clinical, Saudi public-hospital framework integrating discounted TCO, BWM expert weighting, TOPSIS ranking, and sensitivity analysis for high-cost medical-device procurement. The outputs of this study directly close that gap.

The criteria hierarchy answers what should be evaluated. The discounted TCO model answers how lifecycle cost should be structured. The BWM weights answer how expert priorities can be incorporated transparently. The TOPSIS ranking answers how alternatives can be compared. The sensitivity analysis answers whether the preferred option remains robust under uncertainty. The dashboard answers how the findings can be interpreted by decision-makers.

This is the return on investment of the research: it converts a fragmented academic problem into a usable procurement-governance tool.

7. Conclusion and Expected Outcomes

I close this proposal with a precise claim: Saudi public hospitals need a defensible way to compare high-cost medical-device procurement alternatives when acquisition price, lifecycle cost, supplier reliability, service continuity, governance alignment, digital readiness, and sustainability point toward competing decisions. I do not attempt to redesign national procurement policy, audit NUPCO, or evaluate clinical performance. I pursue a narrower and more rigorous task: developing and testing a reproducible decision-support framework that translates lifecycle cost and multi-criteria value into a transparent procurement ranking.

The expected outcome is not a generic recommendation that hospitals should “buy better.” The expected outcome is a specific analytical package: a validated criteria hierarchy, a discounted Total Cost of Ownership model, BWM-derived expert weights, a TOPSIS ranking matrix, a sensitivity-analysis report, a managerial procurement dashboard, and a replication package documenting assumptions, formulas, and calculation logic. These outputs will show whether the preferred procurement alternative changes when decision-makers move from acquisition-price logic to lifecycle-value logic.

I will answer its central research question by linking each stage of the framework to a specific decision need. The criteria hierarchy will define what should be evaluated. The discounted TCO model will structure CAPEX and lifecycle OPEX. The BWM procedure will identify how experts prioritize financial and non-financial criteria. TOPSIS will convert weighted performance into ranked alternatives. Sensitivity analysis will determine whether the ranking is stable, conditionally stable, or assumption-dependent. This sequence closes the loop between the problem statement, the research gap, the methodology, and the expected contribution.

The theoretical contribution lies in integrating Total Cost of Ownership, value-based procurement, and Multi-Criteria Decision Analysis into one non-clinical health-services-management framework. The practical contribution lies in giving hospital administrators, healthcare finance managers, procurement committees, and public-sector stakeholders a more auditable way to distinguish short-term affordability from long-term procurement value. I will not claim verified national savings, real supplier performance, or actual Saudi procurement prices. It will instead produce the decision architecture through which authorized institutions could later test real procurement data.

Within five years, the value of this research would be visible if procurement teams can use lifecycle-cost templates, structured criteria weights, and robustness testing before awarding high-cost device contracts. The field would know more precisely which assumptions drive procurement rankings: maintenance cost, consumables exposure, downtime, supplier reliability, software burden, local service availability, or governance alignment. That knowledge is currently fragmented, and this study will convert it into a usable model.

The project is credible because it fits my analytical background. My prior work on ROI, CAPEX/OPEX logic, infrastructure evaluation, data-based modelling, and financial decision support gives me a direct foundation for applying lifecycle costing and decision analytics to hospital procurement. This proposal is therefore not an abstract exercise; it is a targeted extension of an analytical trajectory I have already begun.

Future research can extend this framework by replacing synthetic scenarios with real hospital procurement data, testing specific device categories, comparing public and private hospital settings, adding supplier-performance records, or integrating clinical outcome indicators where ethically and institutionally permitted. This study provides the foundation for that next stage by building the model first, defining its assumptions, and proving that it can be tested without restricted data.

The central argument is simple: if Saudi healthcare transformation requires financially sustainable and accountable procurement, then high-cost medical-device decisions need more than acquisition-price comparison. I deliver the first step toward that standard by turning lifecycle cost, expert judgment, and procurement value into a transparent decision model that can be examined, challenged, replicated, and improved.

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Appendices

The appendices provide raw instruments, compliance evidence, technical specifications, and reproducibility materials required to verify the methodology. They are not used to extend the argument beyond the main text. All interpretation, analysis, and conclusions remain in the body of the proposal.

Appendix Index

Table A.1 organizes the appendices as compliance evidence, raw instruments, technical specifications, and reproducibility material rather than as a dumping ground for extra analysis.

Table A.1. Appendix Index

Appendix	Document	Function
I	Participant Information Sheet	Ethics / participant information
II	Informed Consent Form	Voluntary consent and data-use authorization
III	Ethical Approval or Ethics Submission Evidence	Ethical clearance documentation
IV	Expert Questionnaire Instrument	Raw data-collection instrument
V	BWM Scoring Instructions and Consistency Screening Protocol	Weighting reproducibility
VI	TCO Model Specification	Lifecycle-cost formula and assumptions
VII	Synthetic Procurement Scenario Templates	Research-generated alternatives
VIII	TOPSIS Calculation Workflow	Ranking reproducibility
IX	Sensitivity Analysis Plan	Robustness testing
X	Project Timeline and Gantt Chart	Implementation schedule
XI	Detailed Budget and Resource Plan	Operational resources
XII	Risk Register and Contingency Plan	Risk control and Plan B
XIII	Reproducibility and Calculation Package	Replication materials
XIV	Source Coverage and Claim Mapping Audit	Evidence traceability
XV	Optional Supplementary Materials	Non-essential materials if permitted

Note. Appendices are limited to raw instruments, ethics/compliance evidence, technical specifications, and reproducibility material. Analytical interpretation remains in the main text. Author's appendix index.

Appendix I. Participant Information Sheet

Purpose: To provide expert participants with clear information before participation.

Required content:

- Research title.
- Researcher name and affiliation.
- Purpose of the study.
- Why the participant has been invited.
- Expected time commitment.
- Description of the expert questionnaire.
- Statement that participation is voluntary.
- Right to withdraw before submission or within the permitted withdrawal window.
- Explanation that no patient data, clinical records, supplier names, confidential prices, internal hospital ledgers, restricted NUPCO files, or employer-sensitive information will be requested.
- Data use, anonymization, storage, and retention statement.
- Researcher contact information.
- Supervisor or ethics-contact placeholder, if required by the university.

Must be referenced in the main text: Chapter 4.5 Data Collection Procedure and Chapter 4.9 Ethical Considerations.

Appendix II. Informed Consent Form

Purpose: To document voluntary, informed, and ethically valid participation.

Required content:

- Confirmation that the participant has read and understood the Participant Information Sheet.
- Confirmation that participation is voluntary.
- Consent to complete the expert questionnaire.
- Confirmation that the participant will not disclose confidential employer, supplier, contract, procurement, patient, or internal institutional data.
- Consent for anonymized responses to be used in the research.
- Consent for aggregated reporting.
- Statement on withdrawal rights.
- Signature or electronic consent checkbox.
- Date of consent.

Recommended format: A one-page consent form using checkbox statements and a final signature/e-consent confirmation.

Must be referenced in the main text: Chapter 4.9 Ethical Considerations.

Appendix III. Ethical Approval or Ethics Submission Evidence

Purpose: To prove that the expert-judgment component has been reviewed or submitted for ethical review.

Required content:

- Final ethics approval letter, if already approved; or ethics submission confirmation, if approval is pending; or ethics application summary, if the proposal is submitted before formal approval.
- Institution or committee name.
- Application title.
- Researcher name.
- Date submitted or approved.
- Approval or reference number, if available.
- Conditions or restrictions imposed by the committee, if any.
- Statement that no data collection will begin before ethics approval, if approval is still pending.

Must be referenced in the main text: Chapter 4.9 Ethical Considerations and Chapter 5.7 Risk Management and Contingency Plan.

Appendix IV. Expert Questionnaire Instrument

Purpose: To provide the raw data-collection instrument used for expert criteria validation and BWM weighting.

Required content:

- Part A - Respondent Profile: role category, years of experience, area of expertise, and exposure to procurement, finance, administration, supply-chain, biomedical equipment, health economics, or public procurement governance.
- Part B - Criteria Validation: relevance rating, clarity rating, measurability rating, overlap/ambiguity flag, and missing-criteria suggestion field.
- Part C - Best-Worst Method Comparisons: 1-9 scale instructions, best criterion selection, worst criterion selection, best-to-others table, others-to-worst table, and completion check.
- Part D - Optional Qualitative Comments: feasibility concerns, missing criteria, implementation barriers, and risks of using the framework in real procurement settings.

Must be referenced in the main text: Chapter 4.5 Data Collection Procedure and Chapter 4.6.2 BWM Weighting Analysis.

Appendix V. BWM Scoring Instructions and Consistency Screening Protocol

Purpose: To make expert weighting reproducible.

Required content:

- Definition of BWM.
- Explanation of best criterion and worst criterion.
- 1-9 preference scale.
- Example of best-to-others comparison.
- Example of others-to-worst comparison.
- Completeness rules.
- Plausibility screening rules.
- Consistency-screening logic.
- Rule for excluding invalid responses.
- Rule for aggregating valid expert weights.

Must be referenced in the main text: Chapter 4.6.2 BWM Weighting Analysis.

Appendix VI. TCO Model Specification

Purpose: To document the lifecycle-cost formula and assumptions used in the synthetic procurement scenarios. This appendix operationalizes the discounted Total Cost of Ownership model used in Section 4.6.3.

The TCO model will use the structure shown in Appendix Equations (A1a)-(A1b).

$$TCO_i = CAPEX_i + \sum_{t=1}^T \frac{LC_{it}}{(1+r)^t} + \frac{DISP_i}{(1+r)^T} - \frac{RV_i}{(1+r)^T} \quad (A1a)$$

$$LC_{it} = OPEX_{it} + M_{it} + C_{it} + S_{it} + D_{it} \quad (A1b)$$

Required assumptions and controls:

- The lifecycle horizon T must be stated in years for each scenario.
- The discount rate r must be treated as a scenario parameter unless an official rate is explicitly justified.
- CAPEX components must include acquisition, installation and facility adaptation where applicable.
- OPEX components must be separated into maintenance, consumables, software or licensing, downtime and disposal-related costs to avoid double counting.
- Low, base and high assumptions must be documented for uncertain parameters such as maintenance cost, consumables cost, downtime cost and residual value.
- Synthetic values must be labelled as illustrative scenario inputs, not as verified Saudi procurement prices or supplier-performance data.
- All units must be stated consistently, preferably in Saudi riyals (SAR) and years.

Must be referenced in the main text: Chapter 4.6.3 Discounted TCO Modelling.

Appendix VII. Synthetic Procurement Scenario Templates

Purpose: To show how the procurement alternatives will be constructed without confidential procurement data.

Required content:

- Alternative A: lowest-CAPEX option.
- Alternative B: lifecycle-efficiency option.
- Alternative C: strong local-service option.
- Optional Alternative D: managed-service or lease-based option.
- Cost assumptions for each alternative.
- Non-price assumptions for each alternative.
- Supplier-risk assumptions.
- Digital-readiness assumptions.
- Sustainability assumptions.
- Clear warning that these scenarios are research-generated and not actual supplier offers.

Must be referenced in the main text: Chapter 4.4 Data Sources, Chapter 4.6.3 Discounted TCO Modelling, and Chapter 5.6 Minimum Viable Thesis Design.

Appendix VIII. TOPSIS Calculation Workflow

Purpose: To make the TOPSIS ranking method auditable and reproducible. This appendix documents the normalization, weighting, ideal-solution, distance and closeness-coefficient calculations used in Section 4.6.4.

The raw decision matrix will be normalized using Equation (A2).

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}} \quad (\text{A2})$$

Where:

- r_{ij} is the normalized value of alternative i on criterion j .
- x_{ij} is the raw value of alternative i on criterion j .
- m is the number of procurement alternatives.
- i indexes alternatives and j indexes criteria.

The weighted normalized matrix will then be calculated using Equation (A3).

$$v_{ij} = w_j r_{ij} \quad (\text{A3})$$

Where:

- v_{ij} is the weighted normalized score of alternative i on criterion j .
- w_j is the BWM-derived weight of criterion j .
- r_{ij} is the normalized value of alternative i on criterion j .

The positive and negative ideal solutions will then be identified for each criterion using Equation (A4).

$$A^+ = \{v_1^+, v_2^+, \dots, v_n^+\}, \quad A^- = \{v_1^-, v_2^-, \dots, v_n^-\} \quad (\text{A4})$$

Where:

- A^+ is the positive ideal solution across all criteria.
- A^- is the negative ideal solution across all criteria.
- v_{j+} is the best weighted normalized value for criterion j .
- v_{j-} is the worst weighted normalized value for criterion j .
- n is the number of decision criteria.

The distance of each alternative from the ideal solutions will then be calculated using Equation (A5).

$$S_i^+ = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^+)^2} \quad (\text{A5a})$$

Where:

- S_{i+} is the distance of alternative i from the positive ideal solution.
- v_{ij} is the weighted normalized score of alternative i on criterion j.
- v_{j+} is the positive ideal value for criterion j.

The distance from the negative ideal solution will be calculated using Equation (A5b).

$$S_i^- = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^-)^2} \quad (\text{A5b})$$

Where:

- S_{i-} is the distance of alternative i from the negative ideal solution.
- v_{ij} is the weighted normalized score of alternative i on criterion j.
- v_{j-} is the negative ideal value for criterion j.

Finally, the closeness coefficient will be calculated using Equation (A6).

$$C_i^* = \frac{S_i^-}{S_i^+ + S_i^-} \quad (\text{A6})$$

Where:

- C_{i*} is the TOPSIS closeness coefficient of alternative i.
- A higher C_{i*} indicates stronger relative closeness to the ideal procurement profile.
- Alternatives will be ranked from highest to lowest C_{i*} .

Must be referenced in the main text: Chapter 4.6.4 TOPSIS Ranking Analysis.

Appendix IX. Sensitivity Analysis Plan

Purpose: To show how ranking robustness will be tested.

Required content:

- Weight sensitivity scenarios.
- Discount-rate scenarios.
- Maintenance-cost scenarios.
- Consumables-cost scenarios.
- Downtime-cost scenarios.
- Supplier-risk scenarios.
- Residual-value scenarios.
- Local-service availability scenarios.
- Rule for classifying results as stable, conditionally stable, or assumption-dependent.

Must be referenced in the main text: Chapter 4.6.5 Sensitivity and Robustness Analysis.

Appendix X. Project Timeline and Gantt Chart

Purpose: To provide the implementation schedule in a visually scannable format.

Required content:

- Month-by-month timeline.
- Work packages.
- Milestones.
- Dependencies.
- Ethics approval window.
- Expert recruitment window.
- Data analysis window.
- Writing and revision window.

Recommended format: Use one landscape-oriented Gantt table if necessary. It must fit on one page or two clean pages maximum.

Must be referenced in the main text: Chapter 5.2 Project Timeline and Milestones and Chapter 5.3 Gantt-Ready Work Plan.

Appendix XI. Detailed Budget and Resource Plan

Purpose: To justify the operational resources required for the project.

Required content:

- Literature access.
- Survey platform.
- Reference management.
- Analysis tools.
- Data storage.
- Recruitment/communication.
- Optional participant appreciation, if ethically approved.
- Contingency reserve.
- Cost-control strategy.
- Total estimated budget range.

Recommended format: Use a single budget table with four columns: budget item, purpose, estimated cost, cost-control strategy.

Must be referenced in the main text: Chapter 5.4 Resource and Budget Requirements.

Appendix XII. Risk Register and Contingency Plan

Purpose: To demonstrate that operational risks have been anticipated before data collection.

Required content:

- Risk description.
- Probability.
- Impact.
- Early warning signal.
- Mitigation.
- Plan B.
- Linked objective affected.
- Minimum risks: ethics approval delay, insufficient expert responses, inconsistent BWM responses, weak public cost data, ranking instability under sensitivity analysis, scope creep, and timeline slippage.

Must be referenced in the main text: Chapter 5.7 Risk Management and Contingency Plan.

Appendix XIII. Reproducibility and Calculation Package

Purpose: To allow another researcher to reproduce the framework.

Required content:

- Variable dictionary.
- Criteria dictionary.
- Assumption log.
- Formula log.
- Spreadsheet/codebook structure.
- Version-control note.
- File-naming convention.
- Data-cleaning rules.
- Aggregation rules.
- Replication checklist.

Must be referenced in the main text: Chapter 4.8 Validity, Reliability, and Reproducibility and Chapter 6.1 Expected Deliverables.

Appendix XIV. Source Coverage and Claim Mapping Audit

Purpose: To show traceability between major claims and supporting source categories.

Required content:

- Claim category.
- Source type.
- Evidence status.
- Use in proposal.
- Limitation or caution.
- Whether the claim is used in Introduction, Literature, Methodology, or Contributions.
- Important restriction: this appendix must not become a second literature review. It is a traceability audit only.

Must be referenced in the main text: Chapter 2.7 Synthesis and Gap Generation or Chapter 4.8 Validity, Reliability, and Reproducibility.

Appendix XV. Optional Supplementary Materials

Purpose: To store non-essential but useful materials only if permitted by the university format.

Required content:

- Allowed: full list of excluded criteria and reason for exclusion; optional VIKOR robustness notes, if used; additional synthetic scenario tables; extended dashboard mock-up.
- Prohibited: new arguments not discussed in the main text; extra literature review paragraphs; admission-positioning material; personal motivation statements; prompt engineering notes; reviewer-style internal checklists; duplicate versions of sections already included in the body.
- Include this appendix only if a specific section refers to it. Otherwise, delete it.

Must be referenced in the main text: Only include this appendix if a specific section refers to it.

Appendix Cross-Reference Requirements

Table A.2 lists the minimum in-text call required for each appendix so that no appendix remains detached from the methodology or implementation plan.

Table A.2. Appendix Cross-Reference Requirements

Appendix	Minimum in-text call
Appendix I	Participant information will be provided before participation (see Appendix I).
Appendix II	Expert participation will require informed consent (see Appendix II).
Appendix III	Data collection will not begin before ethics approval or ethics clearance confirmation (see Appendix III).
Appendix IV	The complete expert questionnaire is provided in Appendix IV.
Appendix V	BWM instructions and consistency-screening rules are detailed in Appendix V.
Appendix VI	The TCO formula and cost assumptions are specified in Appendix VI.
Appendix VII	Synthetic procurement scenarios are described in Appendix VII.
Appendix VIII	The TOPSIS calculation workflow is provided in Appendix VIII.
Appendix IX	Sensitivity-analysis scenarios are listed in Appendix IX.

Appendix	Minimum in-text call
Appendix X	The Gantt-ready implementation plan is provided in Appendix X.
Appendix XI	Detailed budget assumptions are provided in Appendix XI.
Appendix XII	The risk register and contingency plan are provided in Appendix XII.
Appendix XIII	The reproducibility package is described in Appendix XIII.
Appendix XIV	The claim-mapping audit is provided in Appendix XIV.

Note. Each appendix must be explicitly cited in the main text before or where the corresponding instrument, ethical document, formula, or planning tool is discussed. Author's cross-reference checklist.